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Curriculum Vitae – Gravitational Wave Astronomy and Numerical Relativity

Publications

Published and archived

18. Yu-Cun Xie, ***Vaishak Prasad***, "Universal Structure of Horizon Formation in Generic Binary Black Hole Mergers", arXiv preprint, 10 September 2026, gr-qc:2609.11819.
17. **Vaishak Prasad**, "NRHJSur3dq8: a spectral numerical-relativity surrogate family for non-eccentric aligned-spin binary-black-hole waveforms", arXiv preprint, 8 September 2026, gr-qc:2609.09088.
    Single author. Released with the public `nrhjsurrogate` package.
16. **Vaishak Prasad**, *Monica Rincón-Ramírez, Nathan K. Johnson-McDaniel, Eugenio Bianchi, Ish Gupta and B. S. Sathyaprakash*, "Do black hole mergers maximize entropy?", Published 27 August 2026, Int. J. Mod. Phys. D, 2641013..
    Gravity Research Foundation 2026 essay, Honourable Mention.
15. Monica Rincón-Ramírez, *Nathan K. Johnson-McDaniel, Eugenio Bianchi, Ish Gupta,* ***Vaishak Prasad*** *and B. S. Sathyaprakash*, "Maximum Entropy Conjecture for Black Hole Mergers", Published 10 July 2026, Phys. Rev. Lett. 137, 021406., arXiv:2601.22388.
    Devised the numerical relativity method used to test the conjecture. The conjecture is framed in terms of the Bondi mass and angular momentum. Also contributed the horizon-geometry analysis.
14. **Vaishak Prasad**, "Accelerated Time-domain Analysis for Gravitational Wave Astronomy", arXiv preprint, 6 March 2026, gr-qc:2603.06010.
    Single author. Exact time-domain likelihood evaluation at near-FFT cost via the Gohberg–Semencul representation. Released with the `tdanalysis` package.
13. **Vaishak Prasad**, "Multi-Segment Consistency Tests of General Relativity", arXiv preprint, 6 March 2026, gr-qc:2603.05835.
    Single author. Demonstrated on GW250114. To be proposed as a reviewed LVK test of general relativity.
12. **Vaishak Prasad**, *and B. S. Sathyaprakash*, "Are black hole spins truly near-zero?", arXiv preprint, 13 February 2026, gr-qc:2602.12859.
11. Divyajyoti, *Isobel M. Romero-Shaw,* ***Vaishak Prasad****, Kaushik Paul, Chandra Kant Mishra, Prayush Kumar, Akash Maurya, Michael Boyle, Lawrence E. Kidder, Harald P. Pfeiffer, Mark A. Scheel*, "Biased parameter inference of eccentric, spin-precessing binary black holes", Published 12 May 2026, Phys. Rev. D 113, 103022., arXiv:2510.04332.
10. **LVK collaboration**, *incl.* ***Vaishak Prasad***, "Black Hole Spectroscopy and Tests of General Relativity with GW250114", Published 30 January 2026, Phys. Rev. Lett. 136, 041403., arXiv:2509.08099.
    Contributed to the analysis and interpretation of the area-law result and its connection with the consistency test. Also contributed to the ringdown model and the review of results. Editors' Suggestion. Featured in *Physics*.
9. **LVK collaboration**, *incl.* ***Vaishak Prasad***, "GW250114: Testing Hawking's Area Law and the Kerr Nature of Black Holes", Published 10 September 2025, Phys. Rev. Lett. 135, 111403..
    Contributed to the analysis and interpretation of the area-law result and to the ringdown model.
8. **LVK collaboration**, *incl.* ***Vaishak Prasad***, "GW230814: investigation of a loud gravitational-wave signal observed with a single detector", Published 18 June 2026, Astrophys. J. Lett. 1004, L23., arXiv:2509.07348.
    Provided the targeted numerical relativity simulation and contributed to writing the paper.

7. **Vaishak Prasad**, "The Shear at the Common Dynamical Horizon in Binary Black Hole Mergers and its Imprint in their Gravitational Radiation.", Published 28 April 2025, Phys. Rev. D 111, 084070..
6. P. Ajith et al., *incl.* ***Vaishak Prasad***, "The Lunar Gravitational-wave Antenna: Mission Studies and Science Case.", Published 28 January 2025, JCAP 01 (2025) 108..
   Author of the Testing GR section
5. **Vaishak Prasad**, "Tidal deformation of dynamical horizons in binary black hole mergers and its imprint on gravitational radiation.", Published 16 February 2024, Phys. Rev. D 109, 044033., arXiv:2109.01193.
   First to show that the horizons of a binary acquire multipolar deformations (hair) during the inspiral, strongly correlated with the gravitational-wave signal at infinity, which the common horizon inherits at merger and loses by absorbing radiation, with quasinormal decay of its multipole moments (damping rate of the dominant moment within 3% of the quasinormal value).
4. Sashideep Gutti, *Shailesh Kulkarni,* ***Vaishak Prasad***, "Closed timelike curves and energy conditions in regular spacetimes", Published 15 December 2022, Eur. Phys. J. C (2022) 82: 1136..
   Equal contribution
3. **Vaishak Prasad**, *Anshu Gupta, Sukanta Bose, Badri Krishnan*, "Tidal deformation of dynamical horizons in binary black hole mergers.", Published 9 February 2022, Phys. Rev. D 105, 044019., arXiv:2106.02595.
2. **Vaishak Prasad**, *Anshu Gupta, Sukanta Bose, Badri Krishnan, and Erik Schnetter*, "News from Horizons in Binary Black Hole Mergers", Published 16 September 2020, Phys. Rev. Lett. 125, 121101., arXiv:2003.06215.
1. **Vaishak Prasad**, *Rahul Srinivasan, Sashideep Gutti*, "The Chronology Protection Problem in modified Kerr-Newman space-times.", Published 14 January 2019, Phys. Rev. D 99, 024023., arXiv:1803.06666.

Theses and reports

2023 Strong Gravity Physics of Dynamical Horizons in Black Hole Mergers and its Imprint in their Gravitational Radiation. Ph.D. thesis, Jawaharlal Nehru University (IUCAA, Pune). Supervisor: Prof. Sukanta Bose. Submitted version in Shodhganga.

2017 Large Scale Magnetic Fields and the Turbulent Dynamo. Project report, IUCAA-NCRA Graduate School. Supervisor: Prof. Kandaswamy Subramanian.

2014 Turbulent Diffusion. Practice School report, BITS Pilani, at IUCAA, Pune.

In preparation

11. Prayush Kumar, *Aditya Vijaykumar,* ***Vaishak Prasad****, Md Arif Shaikh*, "Cross-Catalog Validation of Numerical Relativity Binary Black Hole Waveforms"., In preparation.
10. **Vaishak Prasad**, *B. S. Sathyaprakash and Abhay Ashtekar*, "Horizon-area increase in GW250114: the quantitative area law test"., In preparation.
9. **Vaishak Prasad**, *Abhay Ashtekar and B. S. Sathyaprakash*, "Assessing intrinsic waveform accuracy using the waveform balance laws"., In preparation.
8. **Vaishak Prasad**, "Waveform systematic measurements using the multi-segment consistency test"., LIGO-P2600044, in preparation.
7. **Vaishak Prasad**, "Parallel Template Bank Generation for Non-Eccentric Precessing Binaries"., LIGO-P2600567, in LIGO-Virgo-KAGRA internal review (P&P).
6. **Vaishak Prasad**, *B. S. Sathyaprakash and Abhay Ashtekar*, "A universal law for the balance-law violation of finite-length gravitational waveforms: post-Newtonian origin and the geometry of precession"..
5. Viviana Caceres, ***Vaishak Prasad***, "Time-domain black-hole spectroscopy with a segmented, GPU-accelerated likelihood"..
4. **Vaishak Prasad**, "Cauchy-characteristic matching via generalized-harmonic evolution on compactified null slices. I. Formulation and the hyperbolicity of exact null slicing"..
3. **Vaishak Prasad**, "Batched likelihood evaluation for nested sampling in gravitational-wave parameter estimation"..
2. **Vaishak Prasad**, "Compile-time performance optimization of numerical relativity simulations in C++ using SpEC"..

1. Priyanka Sinha, *Samanwaya Mukherjee, Prayush Kumar,* ***Vaishak Prasad****, Lawrence Kidder, Harald Pfeiffer, Mark Scheel, Md Shaikh, Michael Boyle*, "Can we twist gravitational waveforms from planar eccentric binaries to emulate generic ones"..